MBA342B – PYTHON FOR DATA ANALYSIS

CIA 3

BANK CUSTOMER CAMPAIGN RESPONSE SYSTEM

A Report submitted in partial fulfillment of the requirements for the degree of Master of Business Administration

By

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BANK CUSTOMER CAMPAIGN RESPONSE SYSTEM

Mini Project Report

1. Problem Statement (Add This Section)

Banks regularly conduct marketing campaigns to promote financial products such as loans, credit cards, and savings schemes. However, identifying customers who are most likely to respond to these campaigns is a challenge.

Banks store large amounts of customer data including personal information, transaction history, and credit records. Analyzing this data manually to determine customer eligibility for marketing campaigns is inefficient and time-consuming.

Therefore, there is a need for an automated system that can manage customer information, track financial activity, calculate credit scores, and predict whether a customer is likely to respond to a marketing campaign.

The objective of this project is to develop a **Bank Customer Campaign Response System** using Python that stores customer data in CSV files, processes financial information, and performs campaign response analysis to assist bank managers in decision-making.

2. Introduction

Banks frequently conduct marketing campaigns to promote financial products such as loans, credit cards, insurance plans, and savings schemes. Identifying customers who are most likely to respond to these campaigns helps banks improve marketing efficiency and increase product adoption.

Managing large volumes of customer information manually is difficult and time-consuming. Therefore, an automated system that can store, manage, and analyze customer data is necessary.

The **Bank Customer Campaign Response System** is a Python-based application designed to manage customer information, banking transactions, credit history, campaign results, and customer support requests.

The system stores data in **CSV files**, which act as a lightweight database. Python's **file handling and CSV modules** are used to read and write data.

The system provides two types of access:

- **Bank Manager Portal** – used to manage customer data and perform campaign analysis.
- **Customer Portal** – used by customers to view their profile, check balance, check credit score, and raise support requests.

This project demonstrates how Python can be used to create a **simple banking Customer Relationship Management (CRM) system**.

3. Objectives

The main objectives of this project are:

- Maintain customer records
- Track banking transactions
- Store credit history information
- Calculate customer credit scores
- Analyze customer eligibility for marketing campaigns
- Maintain campaign prediction results
- Handle customer support requests
- Provide a simple prototype of a banking CRM system

4. Technologies Used

Technology	Purpose
Python	Programming language used to implement the system
CSV Files	Data storage for customer and banking data
File Handling	Reading and writing data from CSV files
Jupyter Notebook	Development and implementation environment

5. System Architecture

The system consists of **two main users**:

5.1 Bank Manager

The bank manager is responsible for managing customer data and performing campaign analysis.

Manager functions include:

- Add customer records
- Record transactions
- Update credit history
- Run campaign analysis
- View customer support requests

5.2 Customer

Customers have limited access to the system. They can perform the following operations:

- View their profile
- Check account balance
- Check credit score
- Check campaign eligibility
- Raise support requests

6. Database Design (CSV Files)

The system uses **CSV files as the database** to store all information.

The project uses **five CSV files**:

- customers.csv
- transactions.csv
- credit_history.csv
- campaign_results.csv
- support_requests.csv

Each file stores different types of data required for the system.

6.1 customers.csv

This file stores **personal information of bank customers**.

Fields

CustomerID, Name, Age, Job, MaritalStatus, Education

Example

101,Arjun,35,Manager,Married,Graduate

102,Riya,28,Engineer,Single,Postgraduate

6.2 transactions.csv

This file stores **customer banking transaction details and account balance**.

Fields

CustomerID, Balance, LastTransaction, TransactionType

Example

101,5000,2000,Deposit

102,3000,500,Withdrawal

6.3 credit_history.csv

This file stores **credit-related information used to calculate credit score**.

Fields

CustomerID, PreviousLoans, RepaymentHistory, MissedPayments

Example

101,1,Good,0

102,3,Poor,2

6.4 campaign_results.csv

This file stores **campaign prediction results** generated by the system.

It identifies whether a customer is likely to respond to marketing campaigns.

Fields

CustomerID, Prediction

Example

101,Likely Subscriber

102,Not Likely Subscriber

6.5 support_requests.csv

This file stores **customer complaints or support requests**.

Fields

RequestID, CustomerID, Issue, Status

Example

1,101,ATM card not working, Open

2,102,Need loan information, Resolved

7. System Modules

7.1 Customer Module

The customer module allows customers to access their personal banking information.

Customers can perform the following operations:

- View profile information
- Check account balance
- Check credit score
- Check campaign eligibility
- Raise support requests

7.2 Manager Module

The manager module allows bank employees to manage system data and perform analysis.

Managers can perform the following operations:

- Add new customer records
- Record customer transactions
- Update credit history
- Predict credit score
- Run campaign analysis
- View customer support requests

8. Credit Score Logic

The system calculates a **credit score** for each customer based on their credit history and financial behaviour.

Base Credit Score

600

Score Adjustments

Condition	Score Change
Good repayment history	+80
Balance greater than 3000	+40
Missed payments	-100
More than 2 previous loans	-50

Credit Score Range

300 – 850

9. Campaign Prediction Logic

The system predicts whether a customer is likely to respond to a marketing campaign.

A customer is classified as **Likely Subscriber** if:

- Credit Score > 680
- Balance > 2500

Otherwise, the customer is classified as:

Not Likely Subscriber

The prediction results are stored in **campaign_results.csv**.

10. Menu Structure

Main Menu

BANK CUSTOMER CAMPAIGN RESPONSE SYSTEM

1 Customer Portal

2 Bank Manager Portal

3 Exit

Customer Menu

1 View Profile

2 Check Balance

3 Check Credit Score

4 Check Campaign Eligibility

5 Raise Support Request

6 Back

Manager Menu

1 Add Customer

2 Record Transaction

3 Update Credit History

4 Run Campaign Analysis

5 View Support Requests

6 Back

11. Python Implementation (Jupyter Notebook)

The system is implemented using **Python in Jupyter Notebook**.

Key implementation features include:

- CSV file handling using csv.DictReader and csv.DictWriter
- Modular functions for each operation
- Menu-driven command line interface
- Credit score calculation algorithm

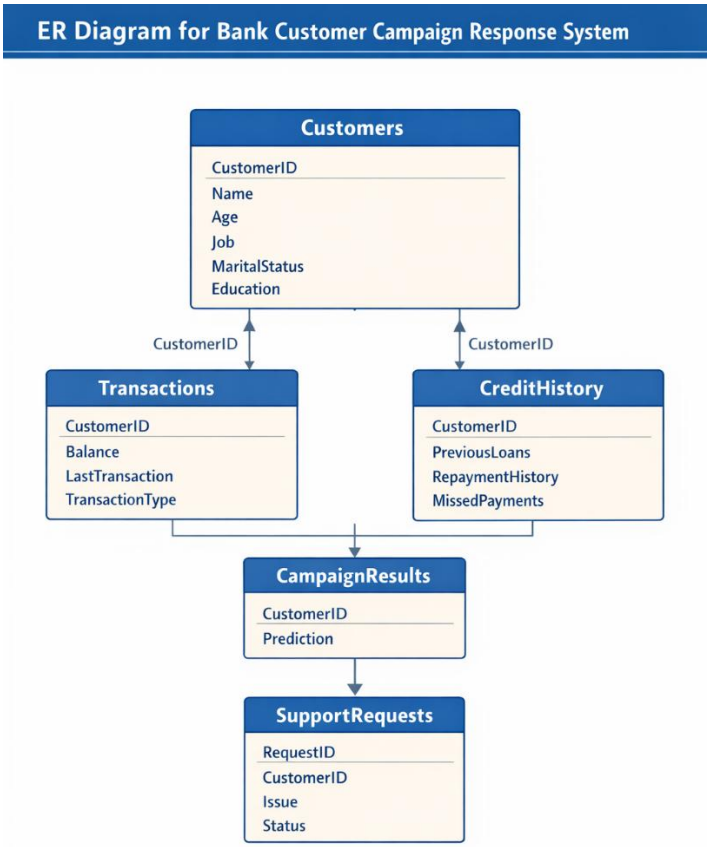
- Campaign prediction logic

Each module is implemented as separate functions within notebook cells.

11. Entity Relationship (ER) Diagram

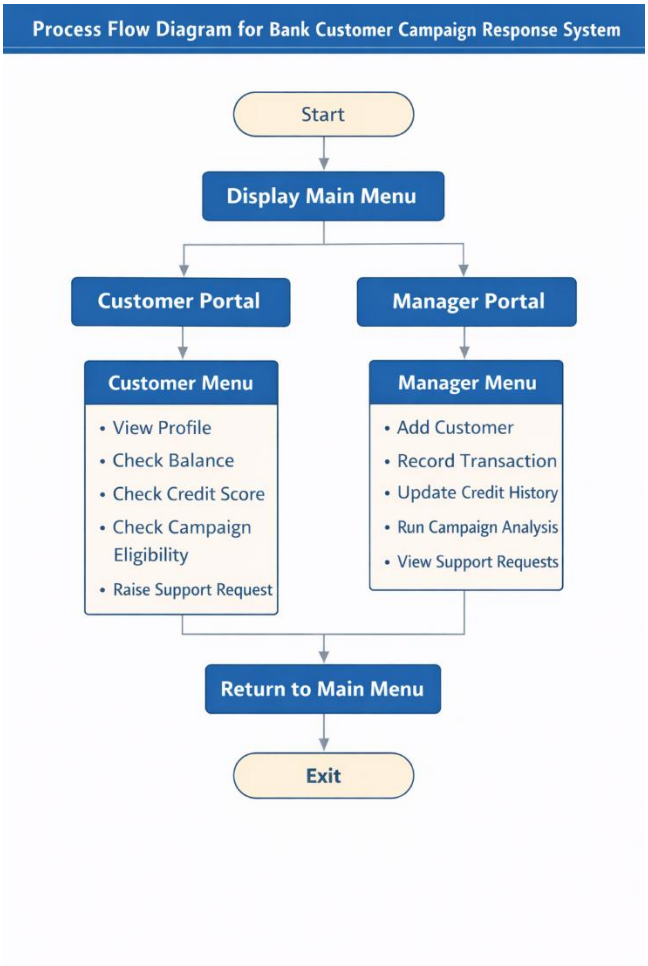
```
        +-------------+
        |  Customers  |
        +-------------+
        | CustomerID  |
        | Name        |
        | Age         |
        | Job         |
        | MaritalStatus|
        | Education   |
        +-------------+
               |
               |
               | CustomerID
               |
   +-----------+-----------+
   |                       |
   v                       v

+-------------+     +-------------+
| Transactions|     | CreditHistory|
+-------------+     +-------------+
| CustomerID  |     | CustomerID  |
| Balance     |     | PreviousLoans|
| LastTransaction|  | RepaymentHistory|
| TransactionType|  | MissedPayments|
+-------------+     +-------------+
```

```
        |
        |
        v

+-------------+
|CampaignResults|
+-------------+
| CustomerID  |
| Prediction  |
+-------------+

        |
        |
        v

+-------------+
|SupportRequests|
+-------------+
| RequestID   |
| CustomerID  |
| Issue       |
| Status      |
+-------------+
```



12. Process Flow Diagram

```
Start
  |
  v
Display Main Menu
  |
  |----------------------|
  |                      |
Customer Portal     Manager Portal
  |                      |
  v                      v
Customer Menu        Manager Menu
  |                      |
View Profile         Add Customer
Check Balance        Record Transaction
Check Credit Score   Update Credit History
Check Campaign       Run Campaign Analysis
Raise Support        View Support Requests
  |                      |
  v                      v
Return to Main Menu
  |
  v
Exit
```



13. Conclusion

The **Bank Customer Campaign Response System** demonstrates how Python can be used to develop a simple banking management application using CSV files as a database.

The system successfully manages customer records, transactions, credit history, campaign predictions, and support requests. It also includes a basic algorithm to predict whether a customer is likely to respond to marketing campaigns.

This project provides practical experience in **Python programming, file handling, data management, and system design**.

In the future, the system can be improved by integrating:

- A graphical user interface (GUI)
- A relational database such as MySQL
- Machine learning models for more accurate campaign prediction

The project serves as a strong foundation for developing more advanced banking information systems.